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New evidence for an allochthonous plate in central southwestern Argentina?

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Abstract—Preliminary paleomagnetic data for rocks from the Choiyoi Formation, exposed at the Cerro Chachil and Sierra de Catan-Lil (39.1°S, 70.5°W, Cordillera Principal, Argentina), assigned *sensu lato* to the Permo-Triassic, are discussed. The directions of the stable remanent magnetization (SRM) for these rocks have been submitted to the fold test. This analysis indicates that most of the Choiyoi Formation rocks are carriers of a prefolding SRM. The mean prefolding SRM for samples of normal and reversed polarity (within 40° of the mean; 277°, +54°, $A_{95}=8°$, N=21, K=15.2, R=19.683) yields a mean pole at 18°S, 126°W, $A_{95}=10°$. The paleomagnetic data indicate two preliminary interpretations. One interpretation suggests that the sampling site rotated 90° clockwise around a vertical axis since Permian times. The other interpretation suggests that the sampling site integrates the allochthonous microplate suggested by Vilas and Valencio (1982) and Valencio and Vilas (1985). This microplate was situated in the Pacific at tropical latitudes in the late Paleozoic.

Resumen—Se discuten los resultados paleomagnéticos preliminares de rocas de la Formacion Choiyoi, aflorante en el Cerro Chachil y Sierra de Catan-Lil (39.1°S, 70.5°O, Cordillera Principal, Argentina), asignada *sensu lato* al Permo-Triásico. Las direcciones de magnetización remanente estable (MRE) de estas rocas fueron sometidas a la prueba de plegamiento. Esta indica que la mayoría de las rocas de la Formación Choiyoi son portadoras de una MRE pretectónica. De dicha MRE promedio para muestras de polaridades normal y reversa (dentro de los 40° de la media; 277°, +54°, $A_{95}=8°$, N=21, K=15.2, R=19.683) se calculó un polo promedio ubicado a los 18°S, 126°O, $A_{95}=10°$. Los resultados paleomagnéticos sugieren dos interpretaciones preliminares. Una de ellas postula que el sitio de muestreo rotó 90° en sentido horario según un eje vertical en tiempos post-Pérmicos. La otra interpretación sugiere que el sitio de muestreo integró la microplaca alóctona sugerida por Vilas y Valencio (1982) y Valencio y Vilas (1985). Esta microplaca habría estado situada en el Pacífico a latitudes tropicales durante el Paleozoico tardío.

INTRODUCTION

THE EXISTENCE of possible allochthonous terranes in southwestern South America has been proposed by several authors (*e.g.*, Nur and Ben Avraham, 1977, 1978; Martinez, 1980; Dalmayrac *et al.*, 1980; Vilas and Valencio, 1982; Ramos, 1984; Ramos *et al.*, 1984; Valencio and Vilas, 1985). In particular, Vilas and Valencio (1982) have suggested two possible interpretations for paleomagnetic data for late Paleozoic rocks from the Cordillera Frontal (Argentina, sampling sites 1 and 3, Fig. 1); one of these interpretations suggests that the section of the Andean Belt located between 31.7°S and 32.1°S was an allochthonous microplate in at least the late Paleozoic. In order to test this interpretation, paleomagnetic studies were carried out in Permo-Triassic volcanic rocks of the Choiyoi Formation exposed at Cerro Chachil and Sierra de Catan-Lil (Argentina, Cordillera Principal: sampling site 2 (39.1°S, 70.5°W), Fig. 1). Preliminary results of these studies are presented in this paper.

GEOLOGIC SETTING

An almost complete and continuous sequence of volcanic rocks of the Choiyoi Formation (Leanza and Leanza, 1979), included in the Choiyoilitense (Groeber, 1946) or Choiyoi Group (Stipanicic, 1965), is exposed at the Cerro Chachil (Cordillera Principal, Argentina, sampling site 2, Fig. 1). The rocks of this sequence, about 800 meters thick, are volcanics of differing composition and thickness. Andesites, dacites, and rhyolites and their pyroclastic equivalents are predominant. Breccias are also intercalated and there are some thin intercalations of epiclastic rocks near the base of the sequence. In this area, the Choiyoi rocks overlie plutonic rocks of the Huechulafquen Formation (Turner, 1965) whose intrusive relationships and radiometric ages indicate a Late Carboniferous to Early Permian age (Di Gregorio and Uliana, 1980; González Díaz and Nullo, 1980). The Choiyoi rocks are uncomformably overlain by a volcaniclastic sequence assigned to the Hettangian-Sinemurian (Chacaico Formation, Parker, 1965; or Lapa Formation, Stipanicic *et al.*, 1968). Therefore, the age of the Choiyoi volcanics is constrained to the Permo-Triassic, *sensu lato*.

The attitude of the Choiyoi rocks at Cerro Chachil is homoclinal (mean azimuth 354°, mean dip 39°); a minor increase of dip at the bottom of the sequence was observed. Ninety-one hand samples, oriented with magnetic compass, were collected from 62 lava flows at this site. Twenty-two oriented hand samples were collected from exposures of 16 lava flows of the same formation at the eastern margin of the Sierra

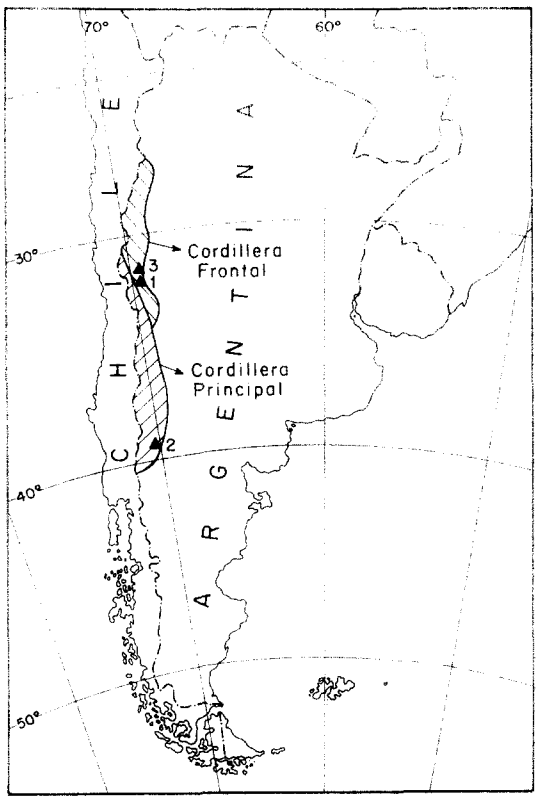


Fig. 1. Sampling sites: 1) Estancia Yalguaraz, Yalguaraz and Portezuelo del Cenizo Formations;; 2) Cerro Chachil and Sierra de Catan-Lil; Choiyoi Formation; 3) Quebrada de la Alumbrera, Choiyoi Group.

de Catan-Lil about 15 km southwest of Cerro Chachil. There, the sampling was carried out at exposures of Choiyoi Formation rocks about 180 meters thick; these rocks probably correlate with those from the middle or the upper section of the complete sequence exposed at Cerro Chachil. The attiude of the Choiyoi rocks at Sierra de Catan-Lil is clearly different from that at Cerro Chachil (mean azimuth 283°, mean dip 53°). This allowed us to apply the fold test (McElhinny, 1964; McFadden and Jones, 1981) to determine if the stable remanence of the Choiyoi Formation rocks was acquired before or after folding.

PALEOMAGNETIC DATA

Nearly one third of the samples collected at Cerro Chachil (pilot samples) and all the samples collected at Sierra de Catan-Lil were submitted to alternating field demagnetization (Valencio, 1980) to analyze the stability of their magnetic remanence and to isolate the stable remanent magnetization (SRM).

SRM was isolated for all the pilot samples from Cerro Chachil; paleomagnetic data for these samples are shown in Table 1. On the other hand, SRM not coincident with the actual dipole was isolated from only eight samples from Sierra de Catan-Lil (Table 1); the others either have a secondary remanence coincident with the direction of the actual dipole at the sampling site or have an unstable remanence.

The directions of the SRM of the lava flows from Cerro Chachil and Sierra de Catan-Lil used in this study were submitted to the fold test. Figure 2a shows that the mean directions of the SRM for these sites, before the structural corrections, are quite different. Figure 2b shows that when the structural corrections are applied these mean directions overlap. This indicates that the SRM isolated from the lava flows from both sites is prefolding.

Virtual geomagnetic poles (VGP) for each lava flow were calculated on basis of their prefolding primary remanence (Table 1). The preliminary mean paleomagnetic pole (PP) for the Choiyoi Formation volcanics from Cerro Chachil and Sierra de Catan-Lil was then computed following Valencio *et al.* (1977): 126°W, 18°S, N = 21, K = 11.6, $A_{95} = 10°$; P14. Table 2 summarizes the late Paleozoic PP's from stable and probably mobile zones of South America used in the following discussion.

DISCUSSION

Figure 3 shows that the position of P14 is far away from the positions of the Permian PP's P5 (263 ± 5 Ma) and P6 (237-297 Ma) from the stable area of

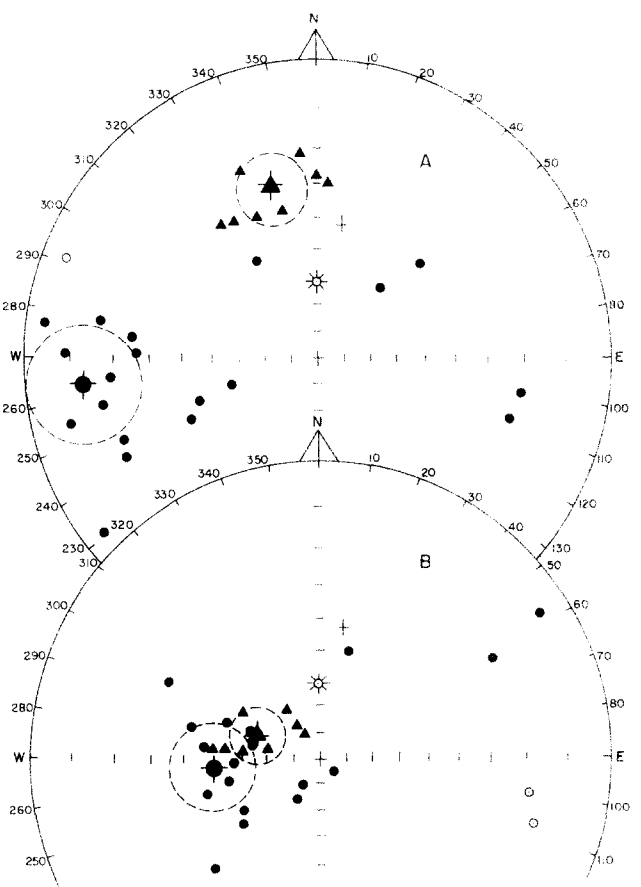


Fig. 2. A) Directions of the stable remanent magnetization (SRM) without structural correction for samples from the Choiyoi Formation: solid circles, from Co. Chachil; solid triangles, from Sa. de Catan-Lil; open symbols, negative inclinations; crossed solid circles, preliminary mean direction of the SRM for Co. Chachil; crossed solid triangles, mean direction of the SRM for Sa. de Catan-Lil. The circles of 95% confidence of the mean SRM are also shown. B) Directions of the SRM after the application of the structural corrections. Present geomagnetic field, +; axial dipole field, spiked circles.

Table 1. Preliminary primary remanence and VGP's for the Choiyoi Formation

Site and Sample No.	Flow No.	JSRM / JNRM	AC (mT)	SRM (in degrees) Inc	SRM (in degrees) Dec	Pre-Folding SRM (in degrees) Inc	Pre-Folding SRM (in degrees) Dec	A_{95}	VGP S	VGP E	A_{95} (in deg.)
Co. Chachil											
38–39	20	0.68	40	8.3	270.7	47.0	273.8	5	14.6	224.1	
40–41	21	0.78	35	15.5	279.4	52.4	288.8	8	7.7	236.2	
43	23	0.99	50	7.1	254.8	45.4	250.9		30.8	210.7	
45	25	0.70	70	2.5	231.1	34.2	223.0		47.8	183.3	*
47	26	0.88	50	16.4	242.2	51.2	229.3		49.7	204.8	
48	27	0.75	70	3.0	277.4	40.7	281.8		5.9	224.8	
49	28	0.35	40	20.6	108.2	−15.3	107.4		8.3	36.6	
50	29	0.80	30	24.7	276.3	62.0	288.4		14.7	243.9	
51	30	1.08	25	55.1	251.5	81.4	140.2		30.9	306.6	*
54	32	0.98	20	42.0	249.2	76.4	209.9		59.4	264.2	*
57	34	0.74	35	18.0	246.8	54.3	235.1		46.4	211.8	
61	38	0.96	20	19.6	100.5	−17.9	100.3		2.1	33.1	
66	42	0.94	75	25.0	271.1	63.4	278.5		21.4	240.7	
67–68	43	0.56	50	15.5	257.8	54.1	253.8	23	32.4	220.0	
9–10	48	1.14	20	38.1	243.9	70.8	208.3	16	65.2	249.2	*
91	49	1.15	70	48.2	327.3	50.2	15.7		−18.4	303.6	*
85	52	0.40	70	18.2	264.5	57.2	264.9		26.2	228.2	
80	55	0.83	55	40.3	48.8	6.3	57.8		−22.2	355.4	*
74	60	0.32	55	55.6	43.6	21.7	60.8		−14.4	351.6	*
71	61	0.98	60	−4.3	291.2	30.0	295.7		−8.5	228.4	
Mean data for site: N = 20				11.4	263.2	50.4	262.8	15	26.3	225.7	17
Sa. Catan-Lil											
6	5	0.57	35	20.8	355.2	67.6	324.4		5.0	267.5	
7	6	0.54	30	21.2	337.6	55.6	300.2		3.3	244.9	
10	8	0.29	25	26.9	359.9	75.0	321.7		15.7	271.6	
15	10	0.39	20	31.8	323.5	49.3	275.2		14.9	226.3	
16	11	0.72	70	35.5	346.4	68.6	278.9		25.1	247.0	
17	12	0.69	70	33.0	327.8	53.1	275.3		16.9	229.3	
21	15	0.50	70	28.9	3.5	78.7	325.8		20.4	276.4	
22	16	0.98	40	34.9	335.3	59.7	275.2		20.9	235.2	
Mean data for site: N = 8				29.9	343.9	64.9	290.8	10	16.0	249.8	14
Mean data for both sites: N = 28				26.0	287.4	55.6	269.4	11	24.1	233.2	13
Mean data for both sites, normal and reversed samples (cutoff 40°): N = 21				20.5	294.5	54.5	277.2	8	17.7	234.0	10

The samples from Co. Chachil are pilot samples. Flows are numbered from the bottom to the top of the sequences. Asterisks indicate VGP with oblique polarity (cutoff 40°; Valencio *et al.*, 1977).

southwestern Mendoza Province (35°S, 68°W). Two preliminary interpretations are proposed to explain these observations.

Interpretation A

The first interpretation postulates that P14 is located on a small circle centered at the sampling site (Δ2, Fig. 3). This suggests a clockwise rotation of the sampling site around a vertical axis centered at this location. Figure 3 shows that when P14 is submitted to a counterclockwise rotation of about 90° along the small circle centered at the sampling site, its position (dashed triangle) is close to the positions of the PP's for the Choiyoi Group — P5 and P6. This suggests that the clockwise rotation of the sampling site was about 90° after Permian times and that the magnetic age of P14 is consistent with those of P5 and P6. On the other hand, the rocks used to compute P14 were collected from two sequences of different attitudes;

this rules out the possibility that the proposed rotation of the sampling site was an apparent rotation produced by non-controlled tectonic movements (MacDonald, 1980).

Interpretation B

The second interpretation postulates that P14 formed part of the late Paleozoic section of the apparent polar wandering curve (APWC) of the accreted microplate presently located in southwestern South America (dotted area, Fig. 4a), proposed by Vilas and Valencio (1982) and Valencio and Vilas (1985). This section of the APWC is also defined by the PP's for the Yalguaraz (C7, middle Carboniferous?) and Portezuelo del Cenizo Formations (P13, Permo-Carboniferous, 229-275 Ma; sampling site 1) and for the Choiyoilitense from Quebrada de la Alumbrera (P12, 203-228 Ma; sampling site 3). Figure 4a shows that this section of the APWC is far away from the late Paleozoic section of the APWC for the stable areas of South America (mainly from the South American platform; Almeida, 1967) defined here by the PP for the Taiguati Formation (C2, Moscovian), the mean of the Late Carboniferous PP's (Cu), and the late Early Permian PP's for the Choiyoilitense (P5 and P6). Paleomagnetic data for these rocks are summarized in Table 2.

Figure 4b shows the relative positions of South America and the allochthonous microplate obtained by the best overlapping of the Middle Carboniferous-late Early Permian sections of their apparent polar wandering curves. The projection is centered at the mean PP (Cu) — that is, the paleolatitudes are valid for the Late Carboniferous (Estephanian). Figure 4b shows that the microplate was located at tropical latitudes in the Pacific during the late Paleozoic. This is consistent with the fusulinid assemblages of the Upper Carboniferous-Lower Permian limestones

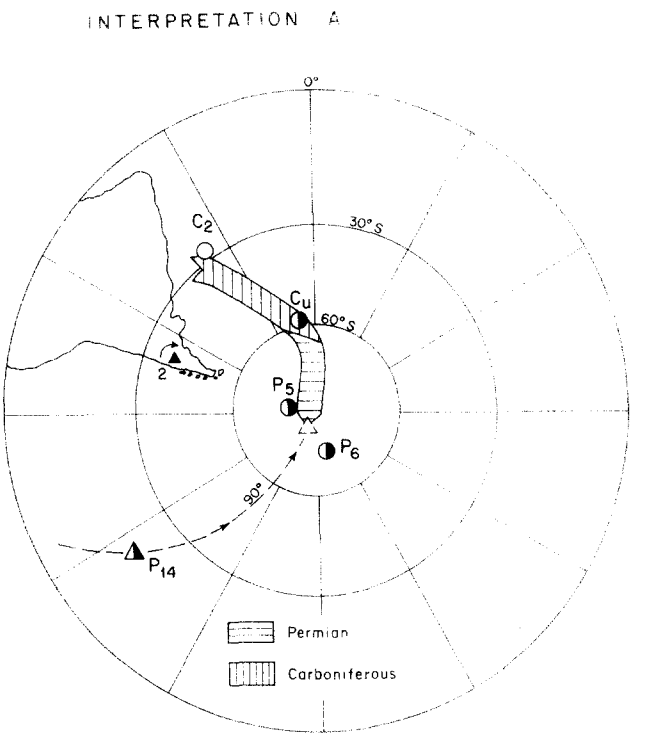


Fig. 3. Interpretation A. Preliminary position of the paleomagnetic pole (PP) for the Choiyoi Formation (P14). The late Paleozoic PP's (circles) and the section of the apparent polar wandering curve (APWC) for stable areas of South America are also shown: - - - →- - -, small circle centered at the sampling site (2); dashed triangle, position of P14 after a counterclockwise rotation of about 90°. The PP's for the stable areas are: C2, Taiguati Formation (Moscovian; Creer, 1965); Cu, Late Carboniferous mean PP (Vilas, 1981); P6, Cochicó Formation (237-297 Ma; Vilas and Valencio, 1982); and P5, Cerro Carrizalito Group (263 ± 5 Ma; Valencio and Vilas 1969). Open symbols indicate normal polarity; solid symbols indicate reversed polarity.

Table 2. South American late Paleozoic paleomagnetic poles used in Figs. 3 and 4.

		Pole Positions				
Formation	**Age**	**Latitude**	**Longitude**	**A_{95}**	**Name**	**References**
Stable Zones:						
Cochico Formation	237-297 Ma	77°S	163°E	5°	P6	Vilas and Valencio, 1982
Cerro Carrizalito Group	263 ± 5 Ma	81°S	282°E	5°	P5	Valencio and Vilas, 1969
South American Late Carboniferous mean pole	Late Carboniferous	58°S	350°E	6°	Cu	Vilas, 1981
Taiguati Formation	Moscovian	28°E	326°E	8°	C2	Creer, 1965
Probably Mobile Zones:						
Choiyoilitense (Qda. Alumbrera, site 3)	203-228 Ma	67°S	290°E	10°	P12	Vilas and Valencio, 1982
Choiyoi Formation	Permo-Triassic	18°S	234°E	10°	P14	This paper
Portezuelo del Cenizo Formation (site 1)	229-275 Ma	32°S	211°E	9°	P13	Vilas and Valencio, 1982
Yalguaraz Formation (site 1)	Mid-Carboniferous?	56°S	195°E	7°	C7	Vilas and Valencio, 1982

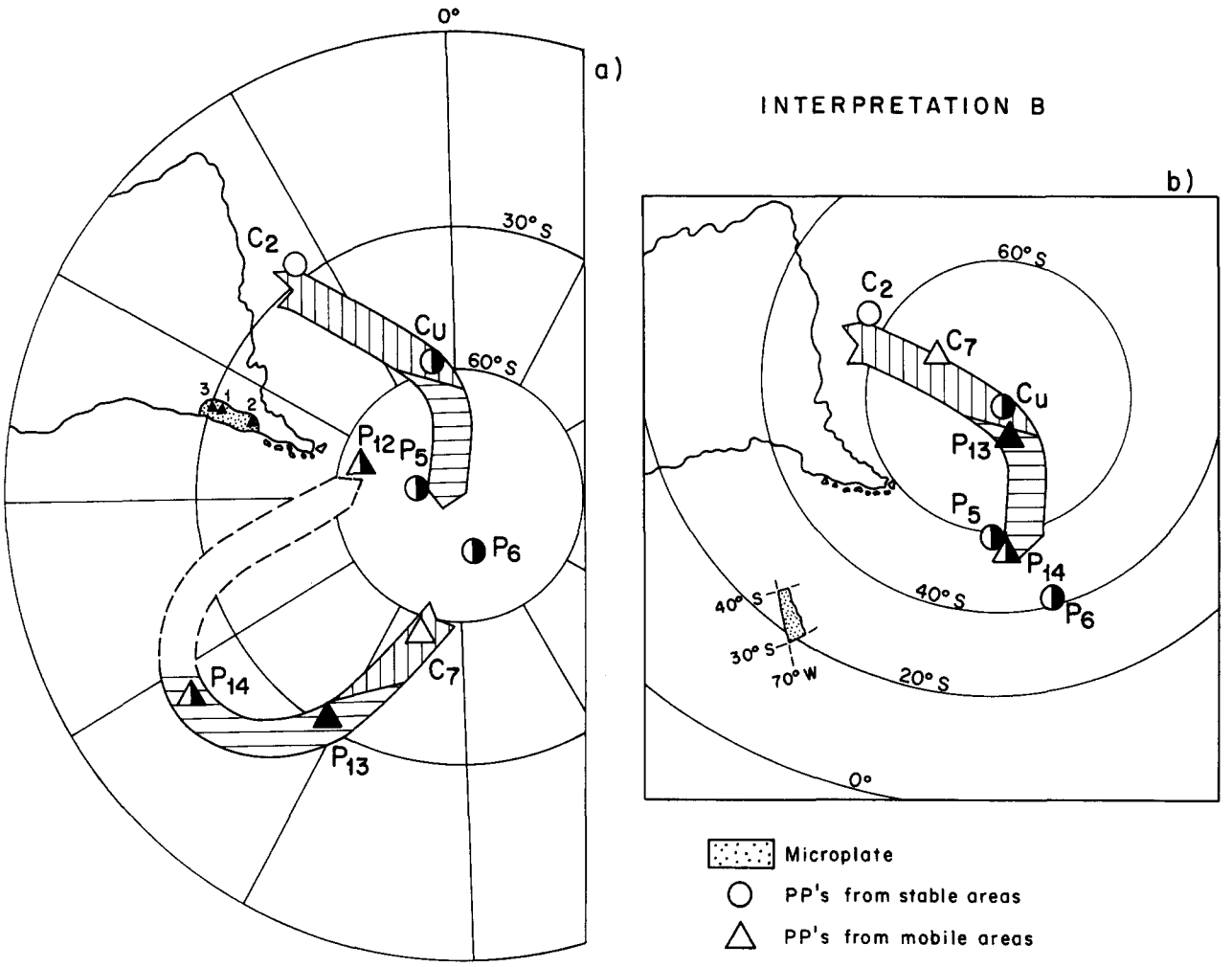


Fig. 4. Interpretation B. a) The sampling sites of the Choiyoi Formation (2), the Yalguaraz and Portezuelo del Cenizo Formations (1), and the Choiyoilitense from Quebrada de la Alumbrera (3) are in an accreted microplate. The PP's for these sites (P14, C7, P13, and P12, respectively) and the late Paleozoic section of the APWC for the microplate defined by them are shown. Also shown is the late Paleozoic section of the APWC for stable areas from South America. References in Fig. 3 and in the text. b) Relative positions of the allochthonous microplate and South America obtained by overlapping of the late Paleozoic sections of their apparent polar wandering curves. Paleolatitudes are valid for the Late Carboniferous.

of the Huentelauquen Formation exposed at 31.4°S, 71.2°W in Chile (Rivano and Sepúlveda, 1983).

According to this model, accretion of the microplate to South America occurred sometime before the time span 228-203 Ma, as suggested by the coincident positions of P12 and the Permo-Triassic PP's from stable zones of South America. In the interests of clarity, these Permo-Triassic PP's are not shown in Fig. 4a.

CONCLUSION

Preliminary paleomagnetic data for Choiyoi Formation rocks exposed at Cerro Chachil and Sierra de Catan-Lil (Andean Belt) suggest two possible interpretations. One of these interpretations suggests a clockwise rotation of about 90° of the sampling site from late Paleozoic times. The other interpretation suggests that the sampling site was part of the allochthonous microplate proposed by Vilas and Valencio (1982) and Valencio and Vilas (1985).

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